

Radiometric Profile Analysis of Soils from the São José de Itaboraí Paleontological Natural Park

J.L. Thalhofer^{1*}, G.R.A. Silva¹, L.S.L. Neto¹, R.L. Silva¹, R.C. Florenzano², T.A. Lima^{1,3}, I.B.M. Ferreira¹, L.B. Silva³, C.O.P. Orejuela³, A.M. Souza³, A.X. Silva³, A. Kelecom¹, V.S. Campos², A.F.M. Freire²,

* e-mail: jardellt@id.uff.br

¹ Laboratory of Radiobiology and Radiometry (LARARA / PPG-DOT / UFF)

² Exploratory Interpretation and Reservoir Characterization Group (GIECAR / PPG-DOT / UFF)

³ Laboratory of Environmental Analysis and Computational Simulation (LAASC / PEN / UFRJ)

G02_015



INTRODUCTION

The Municipal Paleontological Natural Park of São José de Itaboraí was extensively exploited during the last century due to its significant limestone deposits, which provided relevant contributions to major civil engineering works such as the President Costa e Silva Bridge (Rio–Niterói) and the Mário Filho Stadium (Maracanã). During this period of mineral extraction, important paleontological remains were uncovered, including mammals, gastropods, reptiles, and amphibians, with emphasis on the oldest known armadillo and the ancestral form of the rhea, both dated to the Paleocene epoch, approximately 55 million years ago. Fossil records of giant ground sloths from the Pleistocene epoch were also identified.

In 1984, following the cessation of mineral exploitation, a 70-meter-deep depression remained, which was progressively filled with rainwater and groundwater seepage, giving rise to a lake. In 1995, the São José Paleontological Park was formally established and later recognized by the Brazilian Commission of Geological and Paleobiological Sites, an entity associated with the United Nations Educational, Scientific and Cultural Organization, as part of humanity's heritage. At present, the park is open to public visitation, offering various trails and ecotourism activities.

Therefore, given its paleontological relevance and historical–social context, this site was selected for soil radiometric analysis.



Figure 1 – Current situation of the Itaboraí Basin, with site images. 1 and 2) northern margin of the basin; 3) eastern margin of the basin; 4 and 5) São José Fault; and 6) western margin of the basin. Source: Bergqvist et al., 2008.

METHODOLOGY

Soil sampling was carried out at the northern margin of the basin, at point 1. For sample collection, an area with scarce vegetation and root presence was selected. The soil sample was collected at a depth of 10 cm below the surface, in the vicinity of the lake. After collection, the sample was processed at LARARA, where it was homogenized, crushed, and individually stored in polypropylene containers for a minimum period of 28 days to ensure the establishment of secular equilibrium.

METHODOLOGY

Subsequently, the samples were analyzed using an HPGe gamma-ray spectrometer for 28,800 seconds in order to acquire the gamma emission spectrum, enabling the quantification of ²²⁶Ra, ²²⁸Ra, ⁴⁰K, ²³⁸U, and ²³²Th.

The calculation of activity concentration (AC), expressed in Bq·kg⁻¹, for ²²⁶Ra, ²²⁸Ra, ⁴⁰K, ²³⁸U, and ²³²Th was performed through the analysis of 13 selected photopeaks. For ²³²Th, the photopeak corresponding to ²⁰⁸Tl at 583 keV was used; for ²²⁶Ra, the photopeak of ²¹⁴Bi at 609.3 keV; for ²²⁸Ra, the photopeak of ²²⁸Ac at 969.1 keV; for ⁴⁰K, the energy of 1,460.8 keV; and for ²³⁸U, the photopeak of ²¹⁴Bi at 1,764.5 keV.

RESULTS

The results demonstrate that the soil exhibits higher concentrations of ²³²Th and its progeny (²²⁸Ra) when compared to ²³⁸U and its progeny (²²⁶Ra). Furthermore, the radionuclide with the highest activity concentration is ⁴⁰K, as presented in Table 1.

The observed variations in activity concentration between ²³²Th and ²²⁸Ra, as well as between ²³⁸U and ²²⁶Ra, can be attributed to the higher geochemical mobility of radium in comparison to thorium and uranium. In addition, the geological formation strongly influenced the deposition profile, since the occurrence and accumulation of these radionuclides depended on whether the depositional processes were of continental or marine origin.

Table 1 – Activity concentration in soil from the northern margin.

Sample	AC (Bq.Kg ⁻¹)				
	²²⁶ Ra	²²⁸ Ra	⁴⁰ K	²³⁸ U	²³² Th
Solo	9,87	12,03	1.463,45	8,74	17,53

CONCLUSIONS

Thus, in a preliminary results, it can be demonstrated that the Paleontological Park area is of particular relevance for the study of radionuclide levels, as it enables the understanding of radionuclide deposition processes throughout different geological periods in the region.

Furthermore, it is necessary to evaluate the soil-to-plant transfer profile and to perform a broader characterization of additional sampling points within the São José de Itaboraí Paleontological Park. Such efforts are essential for the purpose of environmental monitoring of the entire park, as well as for conducting a more accurate analysis of the soil and bedrock present in the region.

Acknowledgment



References

